

Application Deployment – Sudir pandian

(Deploy the given React application to a production ready state)

Application:

Clone the below mentioned repository and deploy the application (Run application in port 3000).

Repo URL : <https://github.com/Vennilavanguvi/Trend.git>

Docker:

- Dockerize the application by creating Dockerfile
- Build an application and check output using docker image.

Terraform:

- Define infrastructure in main.tf to create VPC, IAM, EC2 with Jenkins, etc.
- Use terraform command to provision infrastructure.

DockerHub:

- Create a DockerHub repository.

Kubernetes:

- Setup Kubernetes in AWS EKS and Confirm EKS cluster is running.
- Write deployment and service YAML files.
- Deploy using kubectl via Jenkins.

Version Control:

- Push the codebase to a Git provider (GitHub).
- Add gitignore and dockerignore files and use CLI commands to push code.

Jenkins:

- Install Jenkins and necessary plugins (Docker, Git, Kubernetes, Pipeline) for build, push & deploy applications.

- Setup Github and jenkins integration using github webhook build trigger for auto build for every commit.
- Create a declarative pipeline script and pipeline project to build, push & deploy using CI-CD.

Monitoring:

- Setup a monitoring system to check the health of the cluster or application (opensource) is highly appreciable.

Submission Guidelines:

- GitHub Link: Submit full code repository.
- README File: Include setup instructions, pipeline explanation, and screenshots or create a document and attach all screenshots.
- Application deployed kubernetes Loadbalancer ARN.

React Application — Production Deployment

DevOps Project Submission Document | Author: Sudirpandian | Date: June 2026

GitHub Repository: <https://github.com/sudirpandian/Devops/tree/project-2> | **Branch:** project-2

1. Project Overview

This document covers the end-to-end production deployment of the Trend React application, including containerization, infrastructure provisioning, Kubernetes deployment, CI/CD pipeline, and monitoring.

Component	Technology	Details
Application	React (Vite build)	Served via Nginx on port 3000
Container	Docker	sudirpandian01/trend-app
Infrastructure	Terraform	AWS VPC, EC2, Security Group, IAM
Kubernetes	AWS EKS	trend-eks, ap-south-1, 2 nodes
CI/CD	Jenkins	Declarative pipeline with GitHub webhook
Monitoring	Prometheus + Grafana	kube-prometheus-stack via Helm

2. Application Setup

2.1 Clone Repository

```
git clone https://github.com/Vennilavanguvi/Trend.git
cd Trend
ls -la
```

```
[root@ip-172-31-11-132 project]# tree Trend/
```

```
Trend/
├── README.md
└── dist
    ├── assets
    │   ├── About-DbHf8GFx.js
    │   ├── Cart-CKKAW61b.js
    │   ├── CartTotal-4jDB2azO.js
    │   ├── Collection-CSvJy8r8.js
    │   ├── Contact-BomtQVpG.js
    │   ├── ForgotPassword-B8rsrKs1.js
    │   ├── Home-Bjlx AQU1.js
    │   ├── Login-B3PL5z7p.js
    │   ├── NewsletterBox-h8A4Pb2y.js
    │   ├── Orders-CaCJbx18.js
    │   ├── PlaceOrder-D9bDxOFB.js
    │   ├── Product-gEYDQbwF.js
    │   ├── ProductItem-CABqN4-8.js
    │   ├── Profile-C4iJY7-d.js
    │   ├── ResetPassword-Dp6S-2CI.js
    │   ├── Title-D4NmpLxE.js
    │   ├── WishList-D8fx7B72.js
    │   ├── about_img-BAJyTXw9.png
    │   ├── contact_img-CyOum2vk.png
    │   ├── hero_img-D0COb6wn.png
    │   ├── index-B52p61-W.css
    │   ├── index-BjbOuBLy.js
    │   ├── logo-BDGV9g83.png
    │   ├── p_img1-BTuXsnJ1.png
    │   ├── p_img10-Bw8q3bg_.png
    │   ├── p_img11-BSK5z8O8.png
    │   ├── p_img12-CWT58dgm.png
    │   ├── p_img13-9keE2fyF.png
    │   ├── p_img14-3-oykR1l.png
    │   ├── p_img15-CEqmdW8L.png
    │   ├── p_img16-DpIAWCPe.png
    │   ├── p_img17-D3TjESyL.png
    │   ├── p_img18-CSUsXTNh.png
    │   ├── p_img19-Ddd8N1Sm.png
    │   ├── p_img20-BN74rvum.png
    │   ├── p_img21-BhDxXaVx.png
    │   ├── p_img22-DIBExxK1.png
    │   ├── p_img23-CGTC-19z.png
    │   ├── p_img24-DxfyHthA.png
    │   ├── p_img25-DpcquDQp.png
    │   ├── p_img26-DAHi2sy3.png
    │   ├── p_img28-CJDN8kul.png
    │   └── p_img2_1-ku4Fyxmj.png
    └──
```

Emoji

Undo

Cut

Copy

Paste

Paste as plain text

Select all

Writing direction

Visual Search

More tools

Inspect

```
├── p_img48-DwPRHR8.png
├── p_img49-CghJxhG1.png
├── p_img5-BMHJXi1.png
├── p_img50-Cg-7CHi5.png
├── p_img51-FROODGKA.png
├── p_img52-CY-YbGLu.png
├── p_img6-B2VXR-Iw.png
├── p_img7-Ds_MlPCY.png
├── p_img8-BeUACWw0.png
├── p_img9-DeR0Ahy1.png
├── razorpay_logo-DrY6y6Wi.png
├── index.html
└── vite.svg
```

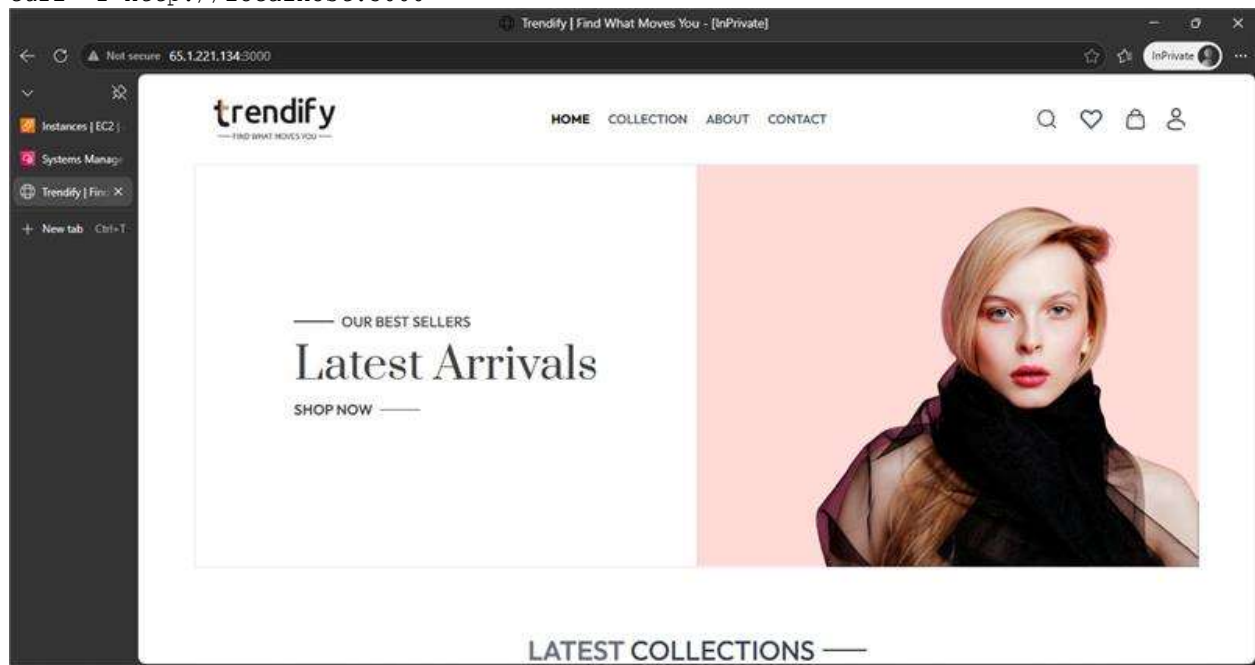
```
directories, 74 files
```

```
root@ip-172-31-11-132 project]# cd Trend/
```

The repository contains a pre-built Vite/React application with only a `dist/` folder — no Node.js source. Static files are served directly via Nginx.

2.2 Run Application Locally (Port 3000)

```
docker run -d -p 3000:3000 trend-app:v1  
curl -I http://localhost:3000
```



3. Docker

3.1 Dockerfile

```
FROM nginx:alpine
COPY dist/ /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 3000
CMD ["nginx", "-g", "daemon off;"]
```

3.2 nginx.conf

```
server {
    listen 3000;
    server_name _;
    root /usr/share/nginx/html;
    index index.html;
    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

3.3 Build and Run

```
docker build -t trend-app:v1 .
docker run -d -p 3000:3000 trend-app:v1
docker ps
```

```
root@ip-172-31-11-132 trend# docker build -t trend-app:v1 .
[+] Building 4.0s (8/8) FINISHED
    [internal] load build definition from Dockerfile
    => transferring dockerfile: 245B
    [internal] load metadata for docker.io/library/nginx:alpine
    [internal] load .dockerignore
    => transferring context: 2B
    [1/5] FROM docker.io/library/nginx:alpine@sha256:b1e7274a03d8b2c95171cc554396fe29abc811c596e27db910edde621e559a
    => resolve docker.io/library/nginx:alpine@sha256:b1e7274a03d8b2c95171cc554396fe29abc811c596e27db910edde621e559a
    => sha256:b1e7274a03d8b2c95171cc554396fe29abc811c596e27db910edde621e559a 10.50kB / 10.50kB
    => sha256:31db554b34376493f4d4e95d8a8d03bbffaf6297fede32d0f923ce2dd529a 2.50kB / 2.50kB
    => sha256:4d944b9b9a7e2c23fe6f422e489dc288f9e7ced8fe44bb0249aaa3f0440 12.41kB / 12.41kB
    => sha256:6a8ac417961e477b048b7f930145215e3c31cfcc9763135a70a4a5adda577bb 3.94kB / 3.94kB
    => sha256:abaaef6d1626e429b3f1209aa369c0a49562cc06b3e075c006e0f699bba5562 1.89kB / 1.89kB
    => sha256:41f234465d8af23123c7e76a302d22ed62cc02d4a51edca72ee752ef01bfaced 421B / 421B
    => extracting sha256:6a0ac1617861a677b048b7f930145215e3c31cfcc9763135a70a4a5adda577bb
    => sha256:de3b677d80020e7e5852f0c046e2c205f424e6dd6f06042a369ad372257 957B / 957B
    => extracting sha256:abaaef6d1626e429b3f1209aa369c0a49562cc06b3e075c006e0f699bba5562
    => sha256:94d553cf70cab544ec790caba5c164d87b3d413617dbcf2fca440d935c16b04 401B / 401B
    => extracting sha256:41f234465d8af23123c7e76a302d22ed62cc02d4a51edca72ee752ef01bfaced
    => extracting sha256:de3b677d80020e7e5852f0c046e2c205f424e6dd6f06042a369ad372257
    => extracting sha256:94d553cf70cab544ec790caba5c164d87b3d413617dbcf2fca440d935c16b04
    => sha256:a3025f4a4b25725673e2fea2b3b9e2094c0b577d324e74abbf14470b5f1cfd 1.40kB / 1.40kB
    => sha256:fbaed1f7fcbef6a0d591ac601924f1f03232ceet2741fde9e950dce5550caf8 20.20kB / 20.20kB
    => sha256:e65d4bbbb9e15a1fa03d42728aeb60bc655f4eb4d6215a49db77a7d104697 1.21kB / 1.21kB
    => extracting sha256:e65d4bbbb9e15a1fa03d42728aeb60bc655f4eb4d6215a49db77a7d104697
    => extracting sha256:a3025f4a4b25725673e2fea2b3b9e2094c0b577d324e74abbf14470b5f1cfd
    => extracting sha256:fbaed1f7fcbef6a0d591ac601924f1f03232ceet2741fde9e950dce5550caf8
    [internal] load build context
    => transferring context: 9.24kB
    [1/2] COPY dist/ /usr/share/nginx/html
    [2/2] COPY nginx.conf /etc/nginx/conf.d/default.conf
    => exporting to image
    => exporting layers
    => waiting image sha256:923f8c93e10db73a30dce67a1a72ef1d6a972260c7b3b5fd05c23e1a0b49e4
    => naming to docker.io/library/trend-app:v1
root@ip-172-31-11-132 trend# docker run -d -p 3000:3000 trend-app:v1
root@ip-172-31-11-132 trend# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
fe20d5ef9aea	trend-app:v1	"/docker-entrypoint..."	5 hours ago	Up 5 hours	80/tcp, 0.0.0.0:3000->3000/tcp, :::3000->3000/tcp	vibrant_swanson

```
root@ip-172-31-11-132 trend#
```

4. DockerHub

```
docker tag trend-app:v1 sudirpandian01/trend-app:v1
```

```
docker push sudirpandian01/trend-app:v1
```

DockerHub Repository: <https://hub.docker.com/r/sudirpandian01/trend-app>

[Explore](#) / [sudirpandian01](#) / trend-app



sudirpandian01/trend-app
By [sudirpandian01](#) · Updated 13 minutes ago
[Watch](#)
☆ 0 ↓ 67

Overview **Tags**

Sort by **Newest**

TAG		
latest		<code>docker pull sudirpandian01/trend-app:latest</code>
Last pushed 13 minutes by sudirpandian01		
Digest	OS/ARCH	Compressed size
f1a7a8f5093f	linux/amd64	33.41 MB

TAG		
v5		<code>docker pull sudirpandian01/trend-app:v5</code>
Last pushed 14 minutes by sudirpandian01		
Digest	OS/ARCH	Compressed size
f1a7a8f5093f	linux/amd64	33.41 MB

5. Terraform Infrastructure

Infrastructure defined in `terraform/main.tf` provisions the Jenkins server in the default AWS VPC.

Resource	Type	Details
VPC	Data source	vpc-08cd654ebd17167be (default)
Security Group	aws_security_group	Ports 22, 8080, 3000 open
IAM Profile	Data source	Existing profile: sudir
EC2 Instance	aws_instance	t3.medium, Ubuntu, Jenkins pre-installed

```
terraform init
terraform validate
terraform plan
terraform apply -auto-approve
```

[illegible]

6. Version Control

6.1 .gitignore

```
node_modules
.env
terraform.tfstate
terraform.tfstate.backup
.terraform/
*.pem
```

6.2 .dockerignore

```
node_modules
.git
.gitignore
README.md
*.pem
```

6.3 Push to GitHub

```
git init
git add .
git commit -m "Initial commit"
git branch -M project-2
git remote add origin https://github.com/sudirpandian/Devops.git
git push -u origin project-2 --force
```

The screenshot shows a GitHub repository interface for a project named 'project-2'. The repository is currently on the 'project-2' branch, which is 9 commits ahead of and 1 commit behind the 'main' branch. The repository has 13 branches and 0 tags. The file list shows the following files and their commit history:

File	Commit Message	Time Ago
root	Add README with setup instructions	da66ed6 - 19 minutes ago
dist	Project Files	4 months ago
k8s	Add k8s manifests and Jenkinsfile	3 hours ago
terraform	Fix Terraform user_data - Java 21 and correct Jenkins GPG ke...	3 hours ago
.dockerignore	Initial commit - trend-app with Dockerfile and nginx config	5 hours ago
.gitignore	Initial commit - trend-app with Dockerfile and nginx config	5 hours ago
Dockerfile	Initial commit - trend-app with Dockerfile and nginx config	5 hours ago
Jenkinsfile	Add KUBECONFIG to Jenkinsfile	2 hours ago
README.md	Add README with setup instructions	19 minutes ago
nginx.conf	Initial commit - trend-app with Dockerfile and nginx config	5 hours ago

The README file is titled 'Trend App - Production Deployment' and has a section for 'Architecture'.

On the right side of the repository page, there are sections for 'Releases' (No releases published), 'Packages' (No packages published), and 'Contributors' (1 contributor: sudirpandian).

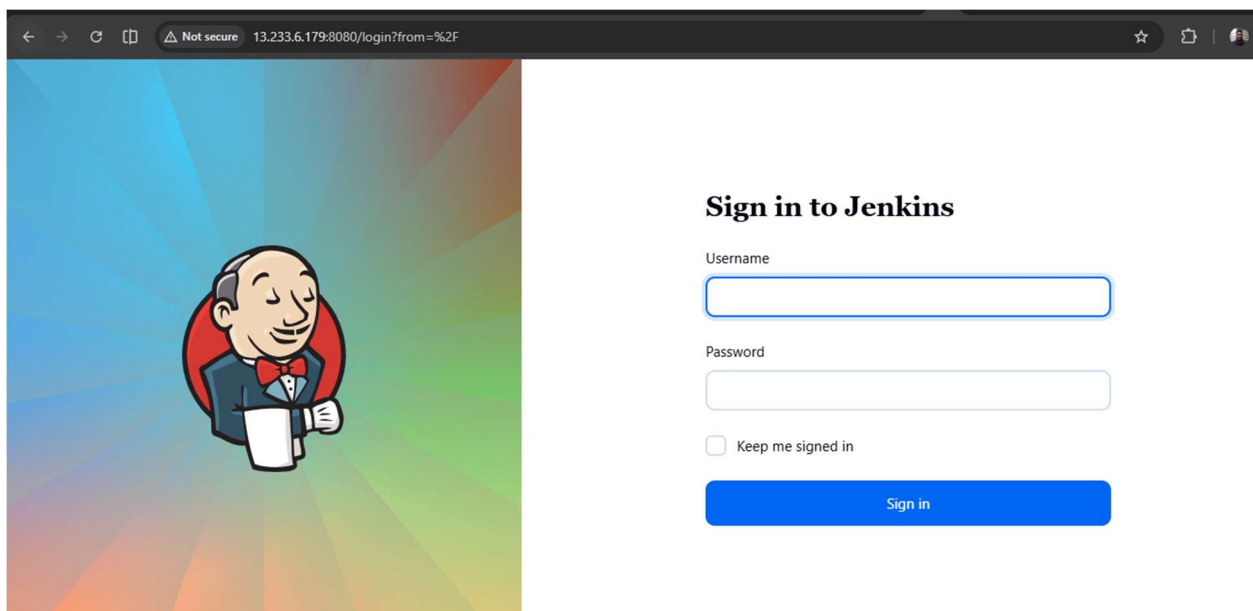
7. Jenkins

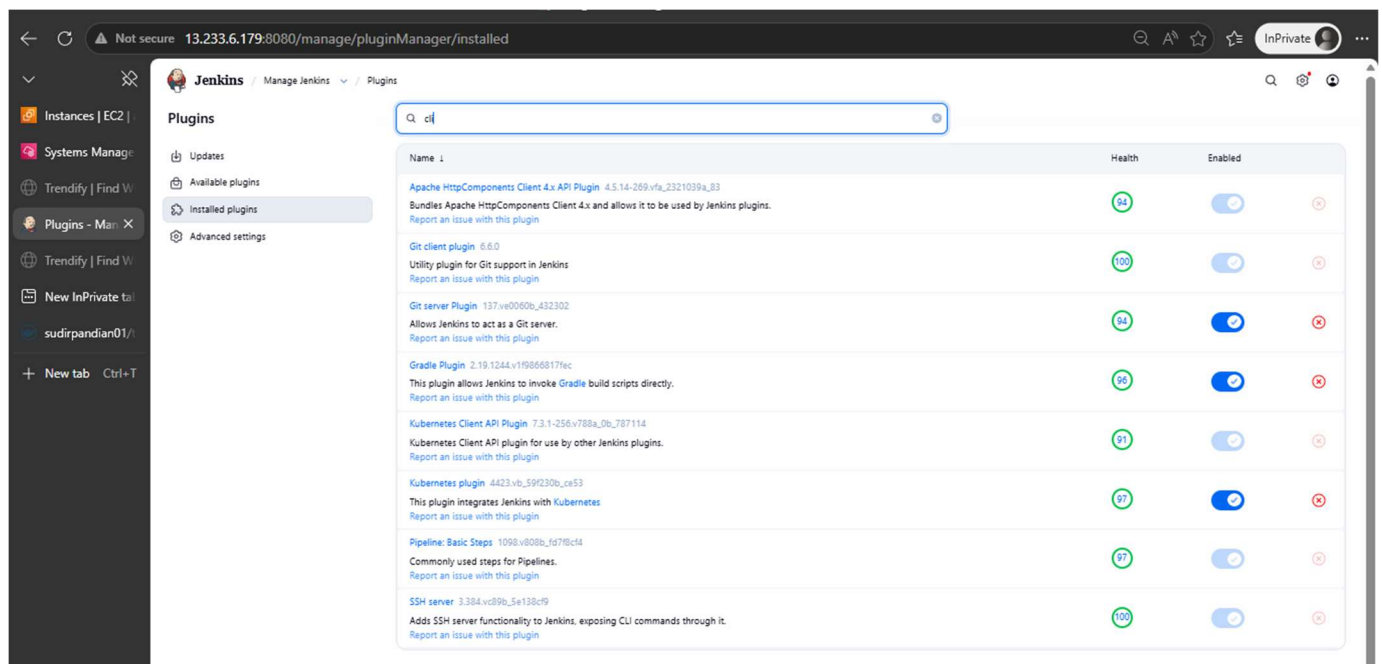
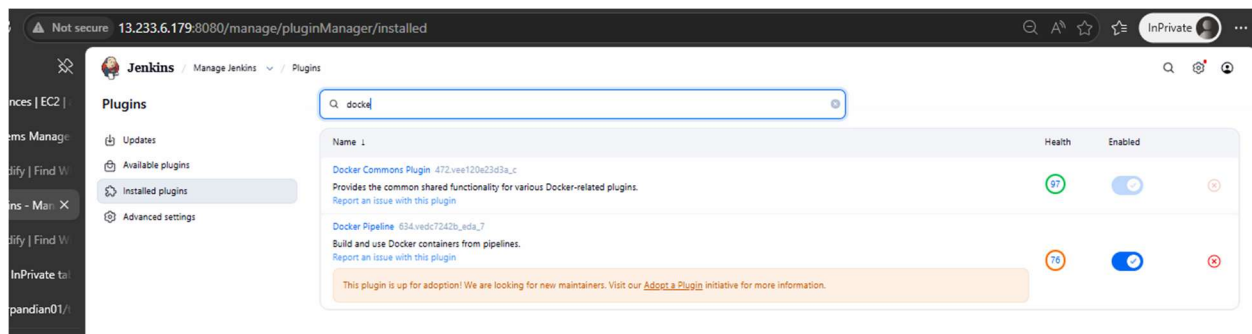
7.1 Installation

Jenkins is provisioned automatically via Terraform user_data on the EC2 instance at **13.233.6.179:8080**. Java 21 (OpenJDK) is required for Jenkins 2.555+.

7.2 Plugins Installed

Plugin	Purpose
Git	Source code checkout
Pipeline	Declarative pipeline support
Docker Pipeline	Docker build and push in pipeline
Kubernetes	Kubernetes integration
Kubernetes CLI	kubectl commands in pipeline
GitHub	GitHub integration
Credentials Binding	Secure credential injection





7.3 GitHub Webhook

Webhook configured at GitHub → Settings → Webhooks:

Payload URL: `http://13.233.6.179:8080/github-webhook/`
 Content type: `application/json`
 Trigger: Push events

github.com/sudirpandian/Devops/settings/hooks/639714552?tab=deliveries

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models

Webhooks

Copilot

Environments

Codespaces

Pages

Security and quality

Advanced Security

Deploy keys

Secrets and variables

Integrations

GitHub Apps

Email notifications

Settings

Recent Deliveries

✓

e12e6d68-6594-11f1-8cf7-f244ea086c38

push

2026-06-11 18:25:54

Request

Response 200

Redeliver

Completed in 0.96 seconds

Headers

Request URL: http://13.233.6.179:8080/github-webhook/

Request method: POST

Accept: */*

Content-Type: application/json

User-Agent: GitHub-Hookshot/576ea10

X-Github-Delivery: e12e6d68-6594-11f1-8cf7-f244ea086c38

X-Github-Event: push

X-Github-Hook-Id: 639714552

X-Github-Hook-Installation-Target-Id: 1005132892

X-Github-Hook-Installation-Target-Type: repository

Payload

{
 "ref": "refs/heads/project-1",
 "before": "74638cde34b0b418ca084357f00be9243f1bf6b7",
 "after": "756e8a91eb2fc4e1ef376d088714c2383a47c002",
 "repository": {
 "id": 1005132892,
 "node_id": "R_kgD00-kcXA",
 "name": "Devops",
 "full_name": "sudirpandian/Devops",
 "private": false,
 "owner": {

8. CI/CD Pipeline

8.1 Pipeline Explanation

Stage	Action	Details
Checkout	Pull source code	Clones project-2 branch from GitHub
Docker Build	Build image	Tags with build number (v1, v2, v3...)
Docker Push	Push to DockerHub	Pushes versioned tag + latest
Deploy to EKS	Rolling update	kubectl set image + rollout status

8.2 Jenkinsfile

```
pipeline {
  agent any
  environment {
    DOCKERHUB_USER = 'sudirpandian01'
    IMAGE_NAME = 'trend-app'
    IMAGE_TAG = "v${BUILD_NUMBER}"
    EKS_CLUSTER = 'trend-eks'
    AWS_REGION = 'ap-south-1'
    KUBECONFIG = '/var/lib/jenkins/.kube/config'
  }
  stages {
    stage('Checkout') { ... }
    stage('Docker Build') { ... }
    stage('Docker Push') { ... }
    stage('Deploy to EKS') { ... }
  }
}
```

13.233.6.179:8080/job/trend-pipeline/5/stages/selected-node=45

Jenkins

trend-pipeline

#5

Stages

Instances | EC2 |

Systems Manage

Trendify | Find W

#5 - trend-pi

Trendify | Find W

New InPrivate ta

sudirpandian01/

New tab Ctrl+T

Start

Checkout SCM

Checkout

Docker Build

Docker Push

Deploy to EKS

Post Actions

End

Search

Post Actions

71ms

Started 25m ago

Jenkins

26ms

Checkout SCM 1s

Checkout 0.09s

Docker Build 2s

Docker Push 11s

Deploy to EKS 10s

Post Actions 71ms

Deployment successful

Deployment successful

13.233.6.179:8080/job/trend-pipeline/5/console

Jenkins

trend-pipeline

#5

Instances | EC2 |

Systems Manage

Trendify | Find W

Console - #5

Trendify | Find W

New InPrivate ta

sudirpandian01/

New tab Ctrl+T

```
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to EKS)
[Pipeline] sh
+ aws eks update-kubeconfig --region ap-south-1 --name trend-eks
Updated context arn:aws:eks:ap-south-1:645437362580:cluster/trend-eks in /var/lib/jenkins/.kube/config
[Pipeline] sh
+ kubectl set image deployment/trend-app trend-app=sudirpandian01/trend-app:v5
deployment.apps/trend-app image updated
[Pipeline] sh
+ kubectl rollout status deployment/trend-app
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "trend-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "trend-app" rollout to finish: 1 old replicas are pending termination...
deployment "trend-app" successfully rolled out
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment successful!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

13.233.6.179:8080/job/trend-pipeline/GitHubPollLog/

Jenkins

trend-pipeline

GitHub Hook Log

Instances | EC2 |

Systems Manage

Trendify | Find W

Jenkins

Trendify | Find W

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Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

GitHub Hook Log

Builds >

Q Filter

Today

#5 2:44 PM

#4 1:00 PM

#3 12:46 PM

#2 12:34 PM

#1 12:30 PM

Last GitHub Push

Started on Jun 11, 2026, 2:44:51 PM

Started by event from 140.82.115.50 => http://13.233.6.179:8080/github-webhook/ on Thu Jun 11 14:44:51 UTC 2026

Using strategy: Default

[poll] Last Built Revision: Revision 4c06e50e43f88e3e36bdefd76899cd33847e2d (refs/remotes/origin/project-2)

The recommended git tool is: git

using credential github-creds

> git --version # timeout=10

> git --version # 'git version 2.43.0'

using GIT_ASKPASS to set credentials github-authToken

> git ls-remote -h -- https://github.com/sudirpandian/Devops.git # timeout=10

Found 13 remote heads on https://github.com/sudirpandian/Devops.git

[poll] Latest remote head revision on refs/heads/project-2 is: da6de66c9f98678f19e662f3b08eaf21845e3d

Using strategy: Default

[poll] Last Built Revision: Revision 4c06e50e43f88e3e36bdefd76899cd33847e2d (refs/remotes/origin/project-2)

The recommended git tool is: git

No credentials specified

> git --version # timeout=10

> git --version # 'git version 2.43.0'

> git ls-remote -h -- https://github.com/sudirpandian/Devops.git

Found 13 remote heads on https://github.com/sudirpandian/Devops.git

[poll] Latest remote head revision on refs/heads/project-2 is: da6de66c9f98678f19e662f3b08eaf21845e3d

Done. Took 1.1 sec

Changes found

9. Kubernetes — AWS EKS

9.1 Cluster Creation

```
eksctl create cluster \  
  --name trend-eks \  
  --region ap-south-1 \  
  --nodegroup-name workers \  
  --node-type t3.medium \  
  --nodes 2 \  
  --managed
```

```
[root@ip-172-31-11-132 trend]$ eksctl create cluster \  
--name trend-eks \  
--region ap-south-1 \  
--nodegroup-name workers \  
--node-type t3.medium \  
--nodes 2 \  
--managed  
2026-06-11 11:38:03 [i] eksctl version 0.227.0  
2026-06-11 11:38:03 [i] using region ap-south-1  
2026-06-11 11:38:03 [i] setting availability zones to [ap-south-1c ap-south-1a ap-south-1b]  
2026-06-11 11:38:03 [i] subnets for ap-south-1c - public:192.168.0.0/19 private:192.168.96.0/19  
2026-06-11 11:38:03 [i] subnets for ap-south-1a - public:192.168.32.0/19 private:192.168.128.0/19  
2026-06-11 11:38:03 [i] subnets for ap-south-1b - public:192.168.64.0/19 private:192.168.160.0/19  
2026-06-11 11:38:03 [i] nodegroup "workers" will use "" (AmazonLinux2022/1.34)  
2026-06-11 11:38:03 [i] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer  
maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/  
2026-06-11 11:38:03 [i] using Kubernetes version 1.34  
2026-06-11 11:38:03 [i] creating EKS cluster "trend-eks" in "ap-south-1" region with managed nodes  
2026-06-11 11:38:03 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup  
2026-06-11 11:38:03 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=ap-south-1 --cluster=trend-eks'  
2026-06-11 11:38:03 [i] Kubernetes API endpoint access will use default of (publicAccess=true, privateAccess=false) for cluster "trend-eks" in "ap-south-1"  
2026-06-11 11:38:03 [i] CloudWatch logging will not be enabled for cluster "trend-eks" in "ap-south-1"  
2026-06-11 11:38:03 [i] you can enable it with 'eksctl utils updateclusterlogging enable --region=ap-south-1 --cluster=trend-eks' or 'eksctl utils updateclusterlogging enable --region=ap-south-1 --cluster=trend-eks --enable-cw-logs'
```

9.2 Verify Nodes

```
kubectl get nodes
```

```
[root@ip-172-31-11-132 terraform]$ kubectl get nodes  
NAME                                                                 STATUS    ROLES    AGE   VERSION  
ip-192-168-61-106.ap-south-1.compute.internal Ready     <none>   3h26m v1.34.8-eks-3385e9b  
ip-192-168-71-102.ap-south-1.compute.internal Ready     <none>   3h26m v1.34.8-eks-3385e9b  
[root@ip-172-31-11-132 terraform]$
```

9.3 Kubernetes Manifests

Files: k8s/deployment.yaml and k8s/service.yaml

```
kubectl apply -f k8s/  
kubectl get pods  
kubectl get svc
```

```
[root@ip-172-31-11-132 terraform]$ kubectl get pods  
NAME                                READY   STATUS    RESTARTS   AGE  
trend-app-7b56c84f8-4181q           1/1     Running   0           29m  
trend-app-7b56c84f8-t8k6w           1/1     Running   0           29m  
[root@ip-172-31-11-132 terraform]$
```

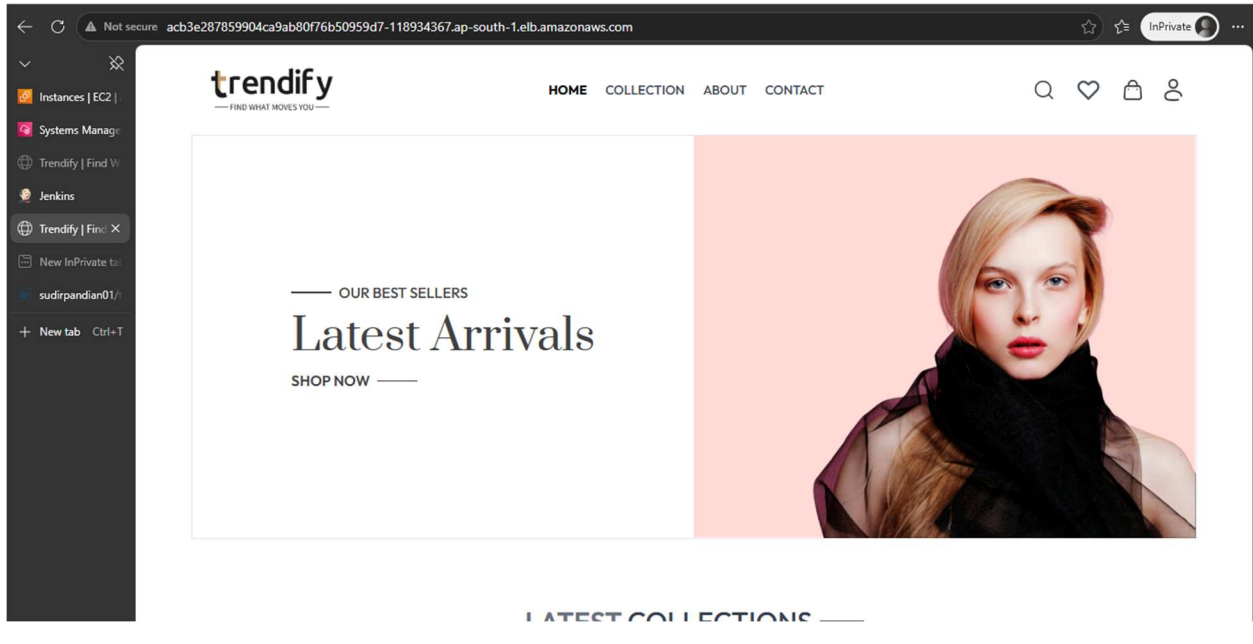
```
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE  
kubernetes    ClusterIP     10.100.0.1       <none>           443/TCP          3h32m  
trend-service LoadBalancer 10.100.164.171   acb3e287859904ca9ab80f76b50959d7-118924267.ap-south-1.elb.amazonaws.com 80:32485/TCP 3h19m  
[root@ip-172-31-11-132 terraform]$
```

10. Application LoadBalancer

LoadBalancer DNS:

acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com

Access URL: <http://acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com>

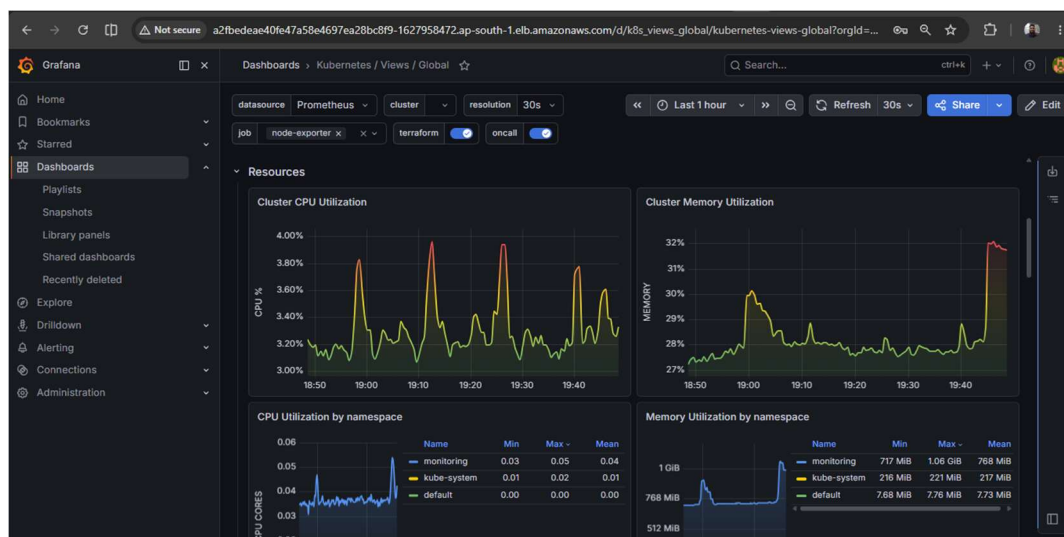
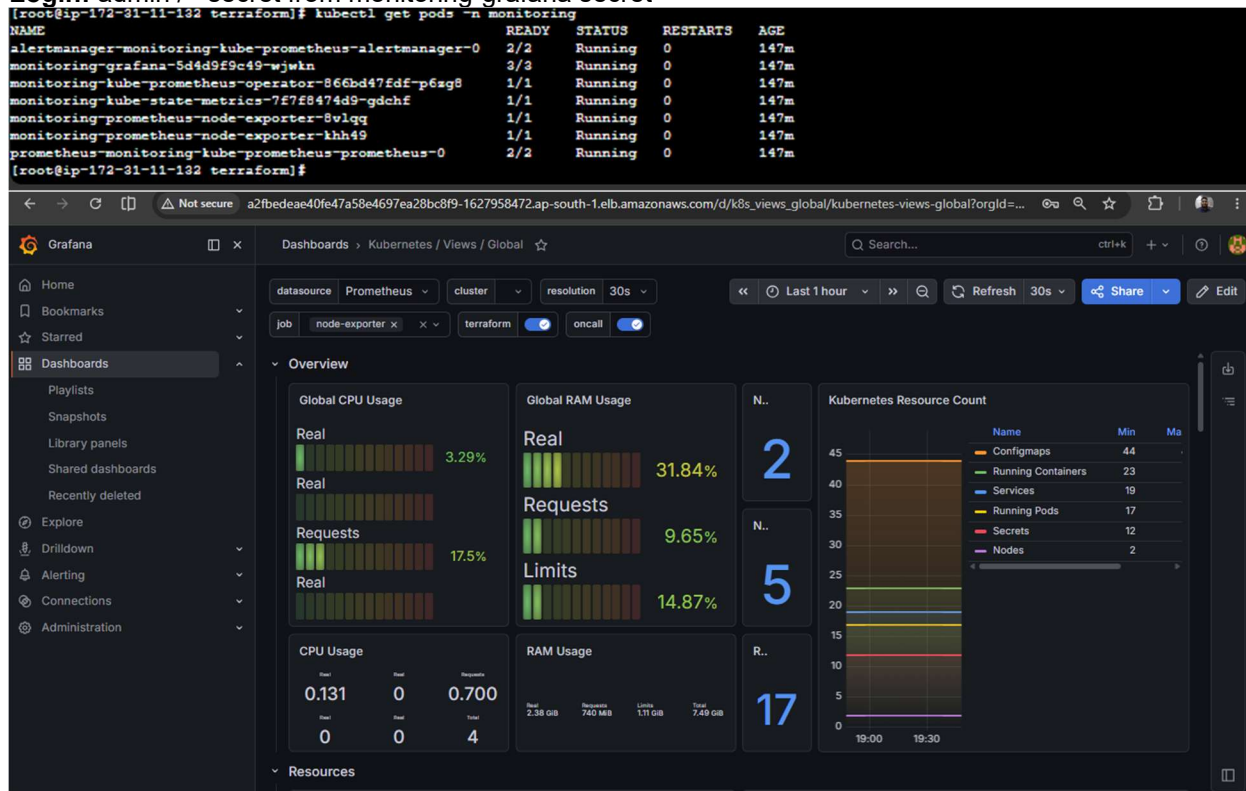


11. Monitoring

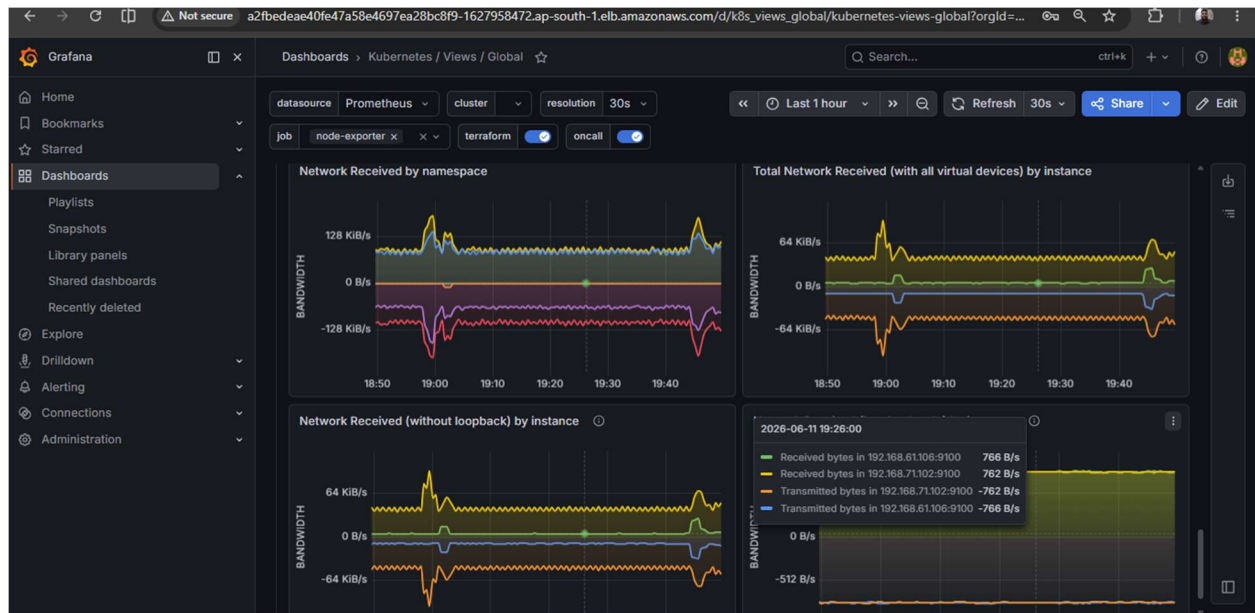
```
helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
helm install monitoring prometheus-community/kube-prometheus-stack \
  --namespace monitoring --create-namespace
kubectl get pods -n monitoring
```

Grafana URL: <http://a2fbedeae40fe47a58e4697ea28bc8f9-1627958472.ap-south-1.elb.amazonaws.com>

Login: admin / <secret from monitoring-grafana secret>







12. Submission Summary

GitHub Repository	https://github.com/sudirpandian/Devops/tree/project-2
DockerHub Image	sudirpandian01/trend-app
Jenkins Server	http://13.233.6.179:8080
EKS Cluster	trend-eks (ap-south-1)
App LoadBalancer ARN	acb3e287859904ca9ab80f76b50959d7-118934367.ap-south-1.elb.amazonaws.com
Grafana	a2fbdeae40fe47a58e4697ea28bc8f9-1627958472.ap-south-1.elb.amazonaws.com